



## Randomized clinical trial evaluating the practical and biological limits of single-feed colostral immunoglobulin G dosing in Australian Holstein and Angus × Holstein calves

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### ABSTRACT

The optimal colostral IgG dose to feed calves remains unclear. We hypothesized that the biologically optimal dose (i.e., the dose that maximizes serum IgG concentration without a plateau in serum IgG response) and the maximum practically feasible dose (the maximum IgG dose that can be reliably delivered due to constraints in colostrum supply) were between 200 and 350 g of IgG. We investigated these hypotheses by comparing serum IgG concentrations in calves block-randomized to feeding groups of either 200 g (n = 45), 250 g (n = 125), 300 g (n = 122), or 350 g (n = 107) of maternal colostral IgG and by monitoring IgG harvest dynamics on a commercial dairy farm in southern Australia. Cows were milked once within 2 h of calving, colostrum was pooled and heat-treated (60°C for 60 min) in batches of 40 L, and calves were administered up to 4 L of colostrum (mean = 3.1 L, range = 1.8 to 3.9) within 3 h of birth. Colostrum samples were titrated into individual dose volumes after calculating IgG concentrations measured using radial immunodiffusion (RID; median = 93 g/L IgG, range = 71 to 121), such that each dose was either 200, 250, 300, or 350 g of IgG (volume = intended dose/measured concentration). Serum IgG was measured using RID on calf serum samples collected at ~2 d of age. A multivariable linear model was used to compare mean serum IgG between colostrum dose groups, controlling for colostrum dose volume. We found that each increasing dose of oral IgG increased serum IgG in a linear fashion between 200 g and 350 g. The most important practical factors limiting IgG dosing were the average IgG yield per cow and the concentration of IgG in colostrum batches. For the farm in this study, the maximum practically feasible dose was 320 g of IgG per calf when feeding a maximum of 4 L colostrum in a single feed. This was based on the find-

ing that the mean IgG from maternal colostrum available per calf during the study was 320 g (118 kg IgG shared among 368 calves) and that most colostrum batches (89%) would be able to deliver 320 g when limited by batch concentration (282–482 g total IgG in batch) and dose ( $\leq 4$  L). We recommend that producers aim to maximize oral IgG intake based on practical feasibility rather than relying on a fixed target minimum dose. It is likely that the maximum practically feasible dose will vary between farms, according to their dynamics of IgG supply, demand, capacity for storage, and delivery.

**Key words:** colostrum, IgG, calves

### INTRODUCTION

Calves are valuable members of the dairy herd because they can generate income via milk and beef production and because they are the vehicle for improving genetic merit of the herd. Disease negatively impacts calf welfare and productivity. For example, calfhood respiratory disease was associated with 0.067 kg lower ADG over the preweaning period and 121.2 kg lower milk production during first lactation (Buczinski et al., 2021). Likewise, Abuelo et al. (2021) observed that heifers with a case of preweaning diarrhea had a 50 g/d reduction in ADG across the preweaning period and also a 325 kg reduction in 305 d mature equivalent production in their first lactation. Other researchers have identified associations between preweaning diarrhea and increased rates of death before first calving, as well as increased rates of departure from the herd during their first lactation (Donovan et al., 1998; Goh et al., 2024).

It is well-recognized that serum IgG is an important factor in disease resistance and productivity in calves (Weaver et al., 2000; Urie et al., 2018b; Godden et al., 2019). In the 1980s, researchers in the United States observed that calves with serum IgG concentrations  $<10$  g/L were at increased risk of mortality (Tyler et al., 1999; Weaver et al., 2000) and thus recommended that colostrum management protocols aim to achieve serum

Received August 26, 2025.

Accepted March 13, 2026.

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The list of standard abbreviations for JDS is available at [adsa.org/jds-abbreviations-26](https://adsa.org/jds-abbreviations-26). Nonstandard abbreviations are available in the Notes.

IgG  $\geq 10$  g/L (Gay, 1983), such that calves with serum IgG  $< 10$  g/L would be considered to have a “failed” transfer of passive immunity and values of 10 g/L or greater would be considered to have been a “successful” transfer. European and American colostrum experts have subsequently identified similar threshold effects for the relationship between serum and mortality (Furman-Fratczak et al., 2011; Todd et al., 2018; Lopez and Heinrichs, 2022). However, a recent large observational longitudinal study (Urie et al., 2018b) of US dairy heifer calves ( $n = 2,545$ ) found an inverse, dose-dependent relationship between increasing serum IgG and decreasing morbidity and mortality. This same dataset was used by colostrum experts in the United States to propose new consensus standards for monitoring passive immunity (Lombard et al., 2020) that reflects this linear relationship. In their paper, they suggest 4 categories of passive immunity: “excellent”  $\geq 25$  g/L, “good” 18 to 24.9 g/L, “fair” 10 to 17.9 g/L, and “poor”  $< 10$  g/L.

Calves acquire serum IgG by ingesting colostrum, a rich source of maternal antibodies absorbed through permeable small intestinal neonatal enterocytes by a process of active pinocytosis during the first 24 h of life (Besser and Gay, 1985; Morin et al., 1997; Barrington and Parish, 2001). The optimal colostral IgG dose to maximize immunity is currently unclear. In the 1990s, feeding a first and single dose of  $\geq 100$  g of IgG was recommended by US researchers to achieve adequate transfer of passive immunity (Besser et al., 1991). Later, recommendations increased to  $\geq 150$  to 200 g of IgG (McGuirk and Collins, 2004; Chigerwe et al., 2008b; Godden et al., 2019), and more recently to 300 g of IgG as a single feed (Godden et al., 2019; Lombard et al., 2020; Robbers et al., 2021b), the latter being the average oral IgG delivered to calves that achieved a serum IgG of  $\geq 25$  g/L (Lombard et al., 2020). The inconsistency in recommendations for target oral IgG doses likely reflects uncertainties regarding the target IgG, the nature of the relationship between oral IgG from colostrum and serum IgG, as well as the relationship with volume and concentration of colostrum fed. The use of a fixed target dose as described above implies that the relationship between oral and serum IgG is curvilinear, such that doses exceeding 300 g would result in insignificant increases in serum IgG and are therefore biologically suboptimal (i.e., diminishing returns theory). This theory is biologically plausible, as absorption of IgG in the gastrointestinal system may be negatively correlated with oral IgG dose due to saturation of the absorptive capacity of the enterocytes. This theory was first introduced by Besser et al. (1991), who found that absorption efficiency started to plateau when calves were administered a single feed of  $\geq 200$  g oral IgG. As dairies intensified and there was increasing need for calves to have higher levels of passive immunity, further

research was performed, but a curvilinear relationship between oral and serum IgG was not observed, forming the general understanding that increasing masses of oral IgG result in linear, incremental increases in calf serum IgG (Hopkins and Quigley, 1997; Godden et al., 2009; Osaka et al., 2014). Findings from other studies have indicated that colostrum volume and concentration may impact absorption efficiency and therefore influence serum IgG concentration (Conneely et al., 2014; Godden et al., 2019), independent of their direct effect on total IgG dose. If IgG absorption does in fact decline with increasing IgG dose, then there is likely to be a “biologically optimal dose,” which maximizes serum IgG concentration without a plateau in the serum IgG response.

It is also important to consider what IgG doses are practically feasible in a commercial farm setting, which is likely to depend on the supply of maternal colostral IgG, calculated as the product of colostrum volume and concentration, the demand (number of calves to dose), and the constraints of administration (maximum of 4 L per calf at a fixed IgG concentration). For example, the total volume that can be administered is limited by the abomasal capacity of the calf and number of feeds that can be administered within 24 h. Godden et al. (2019) state that a calf should be fed 10% to 12% of its BW in colostrum at the first feeding, with newer recommendations suggesting to feed 8% to 10% of calf BW to prevent abdominal discomfort and a reduction in absorption efficiency (Frederick et al., 2025). This, coupled with the assumption that Holstein calves have an average BW of 40 kg, has led to the standard recommended volume of 4 L, as it is rarely practical for farm workers to weigh calves individually (Palczynski et al., 2022). As colostrum is a valuable resource with fluctuating supply, higher quality and volume feeds would historically be prioritized for more valuable stock (Shivley et al., 2019; Palczynski et al., 2020; Renaud et al., 2020). For example, Shivley et al. (2019) reported that bull calves received colostrum at an average of 4.3 h, compared with heifers (mean time to feeding 2.9 h), and received a lower volume of colostrum for subsequent feedings (1.7 L, SD = 1.5) compared with heifer calves (2.5 L, SD = 1.9 L). However, differing colostrum management practices by sex is not always the case (Barry et al., 2019; Robbers et al., 2021a). With the rising value of calves due to dairy beef production, as well as public pressure on the dairy industry to improve calf health and subsequently welfare, there is an increasing demand for colostral IgG to be delivered to all calves to maximize serum IgG. However, with the supply of colostral IgG (i.e., volume and concentration) being subject to breed, parity, season of calving, and nutrition, as well as timing of colostrum harvest due to a dilutional effect with milk production (Denholm et al., 2018; Godden et al., 2019; Mann et al., 2025), further research is

required to determine what IgG dose can be practically delivered to calves.

The primary objective of the study was to identify the biologically optimal dose of colostrum IgG by comparing serum IgG in Holstein and Angus × Holstein (**Angus X**) calves weighing an average of 39 kg, administered 200, 250, 300, and 350 g of IgG. A secondary objective was to identify how colostrum supply (volume and concentration) can impact the practical feasibility of colostrum IgG dosing.

## MATERIALS AND METHODS

A clinical trial was conducted on a dairy farm in southern Australia, milking 3 times daily with an average yield of 42 L/cow per day. Dry and close-up cows were fed a TMR. The farm was purposively selected because it had rigorous calf management protocols and recordkeeping practices and because a sufficient number of calves were due to be born at the time fieldwork was planned. Four hundred and 24 calves were enrolled from October 24 to November 10, 2024.

### Sample Size

Based on work by Osaka et al. (2014) describing the relationship between oral IgG dose and serum IgG ( $y = 0.0804 \times \text{oral dose}$ ), the estimated expected difference in serum IgG among oral doses of 200, 250, and 300 g of IgG, assuming an SD of 5, delta of 0, power of 0.8, and an  $\alpha$  level of 0.05, required a minimum sample size of 70 calves per treatment group. Calculations were conducted using the `epi.sssupb` function in the EpiR package in R (version 2.0.92, Stevenson and Sergeant, 2025).

### Description of Relevant Husbandry Practices on the Study Farm

At the time of the study, cows showing signs of early stage 1 labor (i.e., enlarged udder, raised tail, relaxed pelvic ligaments, or a mucoid vaginal discharge, or all of these; Proudfoot et al., 2013) were brought from close-up compost-bedded pens into small-group calving pens and were monitored for signs of dystocia (e.g., cow not progressing in delivery, calf not presenting in correct orientation). Once born, each calf was left with its dam for up to 30 min before having its navel sprayed with iodine, moved into a group calf pen with wood shavings, and given an ID ear tag in a sequential order. Farm records of birth weight measurements for calves ( $n = 54$ ) born 1 wk after the trial period (i.e., on nonsubjects) indicated that calves weighed 42.1 kg (SD = 6.7) for Angus X calves and 38.9 kg (SD = 5.7) for Holstein calves, making an overall average of 39.6 kg

(SD = 6.01). Individual calf BW were not measured in study calves due to practical constraints.

### Study Subjects: Eligibility Criteria

Calves were included in the trial if their estimated weight was between 35 and 45 kg (based on visual estimate) and they were born without assistance. Animals were excluded retrospectively if they were premature, outside the estimated weight range, experienced dystocia, or if farm staff failed to record the unique colostrum dose ID that was administered. It was more practical to randomize calves by their tag number before birth and, therefore, before their eligibility had been assessed. Consequently, the exclusion process occurred after randomization, however, this was performed without knowing what treatment the calf had been assigned to. Holstein ( $n = 294$ ) and Angus X ( $n = 130$ ) were enrolled. Angus X calves were not originally intended to be enrolled but were included at the request of the farm managers to maintain a consistent routine during the study period.

### Treatment Groups

Calves were allocated (stratified by breed) to receive 200, 250, 300, or 350 g oral IgG within 3 h of birth via esophageal tube. Time of birth and feeding, as well as dose ID of the treatment administered, were recorded by farm staff. Initially, calves were block randomized using the `RAND()` function in Excel (Microsoft Excel, 2024, 2510 edition; block size 6, ratio of 1:1:1) to receive 200, 250, or 300 g of IgG. However, farm managers became concerned when routine monitoring found that the proportion of calves with TP values  $<5.5$  g/L had increased from 6.3% to 22.6% during the first week of the trial. Consequently, no additional calves were enrolled into the lowest dose group (200 g), and the randomization schedule was adjusted to enroll the remaining subjects to doses of 250, 300, and 350 g (i.e., a previously unplanned dose level) at an enrollment ratio of 2:2:3, respectively, using a block size of 7. A study investigator was onsite at the farm to oversee the implementation of the protocols, including the transition to the new treatment group allocations. Staff involved in treatments and collection of samples were blinded to treatment group throughout the study.

**Harvesting and Testing Colostrum.** Within 1 h of calving, cows were administered 1 mL (10 IU) of oxytocin IV (Ilium Syntocin 10 IU/mL, Troy Laboratories Pty Ltd.), received iodine-based premilking teat disinfection (Bovadine Premixed Teat Dip and Spray, Ruakura Pty Ltd.), and had colostrum harvested using a portable milking machine (iMilk 2.0 [Double], Daviesway DAS-CO Pty Ltd.). The volume and the Brix refractometer



(RHB-32 ATC Brix Refractometer, YH Equipment Co. Ltd., Shenzhen City, China) reading (a proxy method to measure IgG concentration of colostrum by refracting light around solids present within a solution; Quigley et al., 2013; Denholm et al., 2022) were recorded for each individual colostrum harvest before placing the colostrum into stainless-steel buckets. The Brix refractometer was calibrated daily using distilled water. Colostrum was refrigerated at 2°C immediately after harvest until enough colostrum was collected to create a pool of 30 to 40 L (maximum storage time 6 h). Colostrum pools were passed through a stainless-steel sieve (to remove teat sealant) into the drum of the pasteurizer machine (Perfect Udder, Dairy Tech Inc.). Each batch of colostrum was heat-treated for 60 min at 60°C and then rapidly cooled to 32°C in line with recent recommendations for reducing bacterial concentrations in pooled colostrum (Johnson et al., 2007; Elizondo-Salazar et al., 2010). After pasteurization, farm staff collected a 30 mL aliquot of colostrum into a clean 50 mL sample jar, which was labeled with the date and batch number, and stored at 2°C for testing by radial immunodiffusion (RID; Bovine IgG Test Kit, Triple J Farms, Bellingham, WA) once daily (see “Measurement of IgG Concentration in Colostrum and Serum” section). The remaining colostrum was stored in 2 L bottles and labeled with the date and a laminated tag on each handle to identify the colostrum batch.

**Preparing IgG Doses.** The heat-treated and refrigerated colostrum was used to create IgG doses (200, 250, 300, and 350 g) based on the batch IgG concentration determined by RID. Each 2 L bottle was mixed by shaking for 10 s. The required volume for each dose was calculated by referring to a dosage table (Supplemental Table S1, see Notes; Hapukotuwa et al., 2025) that showed volumes of each dose for colostrum concentrations ranging from 50 to 200 g/L. The volumes were measured using a 5 L graduated jug. Once each dose was prepared, the bottle(s) were labeled with a colored (blue, pink, or yellow), laminated dose tag that was labeled with dose ID, which traced the dose back to a specific volume and colostrum batch concentration. These volumes were later used to calculate the maximum practically feasible dose (the maximum IgG dose that can be reliably delivered due to constraints in maternal colostrum supply) for the secondary study objective. If multiple bottles were required to create one dose, the handles were zip-tied to indicate pairing. A maximum of 4 L was administered to calves due to the average approximate BW of the study farm calf population being 40 kg and recommendations of feeding 10% BW in a single feeding (Godden et al., 2019; Frederick et al., 2025). Doses were stored in a freezer at −20°C if they were not expected to be used within 24 h.

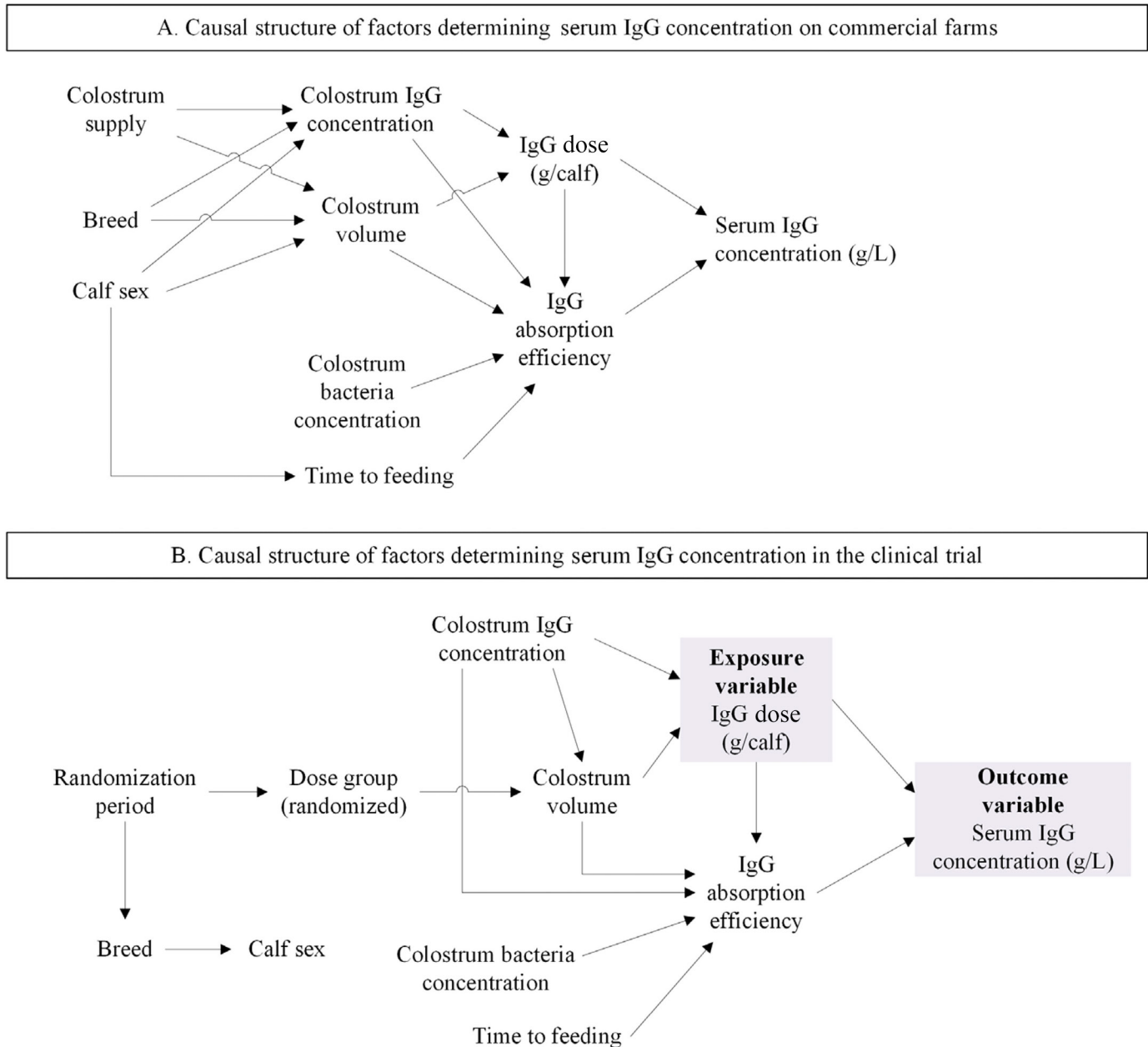
Farm staff used a prelabeled treatment sheet to identify which colostrum treatment group (i.e., label color)

had been randomly assigned to each calf. Samples were retrieved from the refrigerator or freezer and warmed using a warm water bath to 37°C before administration via esophageal tube. Once the calf had been fed, the date, dose ID, volume delivered, time of feeding, and technician name were recorded onto a treatment sheet, along with calving records (including time born, sex, and calving assistance). Note that because the entirety of colostrum supply on-farm was being used to create treatment doses, all animals were administered their assigned treatment dose despite their exclusion (performed without knowledge of their assigned dose) but were subsequently removed from the final analysis. The only exception were the premature calves (defined as a calf being delivered between 230 and 260 d of gestation; Erturk et al., 2025) that were administered 2 L colostrum straight from the pasteurizer machine. These volumes (i.e., nontreatment doses) were still used in the total volume calculations to calculate the maximum practically feasible dose.

**Blood Sampling.** Blood was collected by jugular venipuncture at 2 d of age, as part of the normal farm monitoring program, using a plain serum vacutainer. Once on-farm testing was completed, the remaining serum was collected using a Pasteur pipette and placed into a sterile 5 mL sample vial, labeled with calf ID and stored at −20°C for IgG testing later.

### Measurement of IgG Concentration in Colostrum and Serum

Colostrum and serum concentrations of IgG were tested using RID. Colostrum was tested daily at the farm laboratory, and serum was tested using the same method and test kit at the University of Sydney's Infectious Disease Laboratory in Camden (NSW, Australia). The RID plates were acclimatized to room temperature for 30 min, and samples were thawed to 37°C in a water bath. Reference standards available with the test were 280, 1,400, and 2,800 mg/dL, and were diluted with PBS to create additional values of 700 and 1,800 mg/dL. Reference standards were tested in triplicate, with the repetitions being tested across multiple plates to account for individual plate variance. The results were used to generate a standard curve for each testing batch. Colostrum samples were mixed thoroughly using a vortex machine, diluted based on their Brix reading (1:1, 1:3, 1:5, and 1:7 dilutions for ≤17%, 18%–24%, 25%–29%, and ≥30% Brix, respectively), and tested in triplicate. Serum samples were diluted 1:1 and tested in duplicate. Plates were inoculated with 5 µL per sample well and incubated at 20 to 24°C for 24 h in a moist chamber. Digital calipers were used to measure precipitation ring diameters in duplicate to the nearest 0.1 mm. Ring measurements were averaged, squared, and converted to IgG concentrations us-



**Figure 1.** Causal diagrams (directed acyclic graphs) outlining the causal structure of factors that influence serum IgG concentration on commercial farms (panel A) and within the experimental design of this study (panel B). The causal structure outlined in panel B was used to inform the variables chosen to include in final models. Colostrum IgG concentration and dose volume were initially included as covariates, as previous research has suggested that they may affect IgG absorption and consequently be a source of confounding. Colostral IgG concentration was not included in the final model because of multicollinearity arising from the high correlation between IgG concentration and volume. Time to feeding was not included in the final model because it was not associated with the exposure (i.e., similar across treatment groups). Calf breed and sex were not included in the final model because they were not likely to be associated with the outcome variable. Sensitivity analyses found that inclusion of those variables had limited impact on final effect estimates.

ing the standard curve generated from the reference sera. Reference standards were tested, and a new curve was generated for each day that a RID assay was performed on colostrum ( $n = 20$ ) and calf serum ( $n = 4$ ) to account for test variation between batches.

### Maximum Practically Feasible Dose

A secondary objective was to explore how IgG supply (volume of colostrum and IgG concentration) impacted the practical feasibility of feeding IgG dose targets

**Table 1.** Descriptive statistics, a comparison of treatment groups within a randomized clinical trial in calves receiving 200, 250, 300, or 350 g of colostral IgG within 3 h of birth after exclusions at enrollment (n = 24), losses to follow-up (n = 2) and calves changing treatment group (n = 6)

| Item                                                        | Oral IgG dose (g) group |                   |                   |                   |
|-------------------------------------------------------------|-------------------------|-------------------|-------------------|-------------------|
|                                                             | 200 g (n = 45)          | 250 g (n = 125)   | 300 g (n = 122)   | 350 g (n = 107)   |
| Breed, n (%)                                                |                         |                   |                   |                   |
| Angus × Holstein                                            | 21 (46.7)               | 37 (29.6)         | 42 (34.4)         | 26 (24.3)         |
| Holstein                                                    | 24 (53.3)               | 88 (70.4)         | 80 (65.6)         | 81 (75.7)         |
| Sex, n (%)                                                  |                         |                   |                   |                   |
| Female                                                      | 35 (77.8)               | 98 (78.4)         | 101 (82.8)        | 84 (78.5)         |
| Male                                                        | 10 (22.2)               | 27 (21.6)         | 21 (17.2)         | 23 (21.5)         |
| Interval from birth to feeding (min)                        |                         |                   |                   |                   |
| Mean (SD)                                                   | 56.7 (28.3)             | 61.9 (34.4)       | 57.0 (28.3)       | 61.4 (24.6)       |
| Median [minimum, maximum]                                   | 50.0 [0, 155]           | 50.0 [0, 195]     | 50.0 [0, 150]     | 60.0 [5.00, 130]  |
| Volume of colostrum (L)                                     |                         |                   |                   |                   |
| Mean (SD)                                                   | 2.22 (0.247)            | 2.78 (0.368)      | 3.30 (0.378)      | 3.56 (0.298)      |
| Median [minimum, maximum]                                   | 2.20 [1.80, 2.60]       | 2.70 [2.10, 3.50] | 3.30 [2.50, 3.90] | 3.60 [2.90, 3.90] |
| Treatment group allocation, n (%)                           |                         |                   |                   |                   |
| Randomly allocated                                          | 44 (97.8)               | 123 (98.4)        | 121 (99.2)        | 105 (98.1)        |
| Reallocated after being accidentally administered this dose | 1 (2.2)                 | 2 (1.6)           | 1 (0.8)           | 2 (1.9)           |

over the first 16 d of the trial (October 24, 2024, to November 8, 2024). The last 2 d of the trial period were omitted from the observational period because colostrum batches were not tested using RID (the trial had enough doses in storage for the last 2 d). Consequently, the number of calves included in the analysis for this objective was different from the number enrolled in the IgG treatment trial.

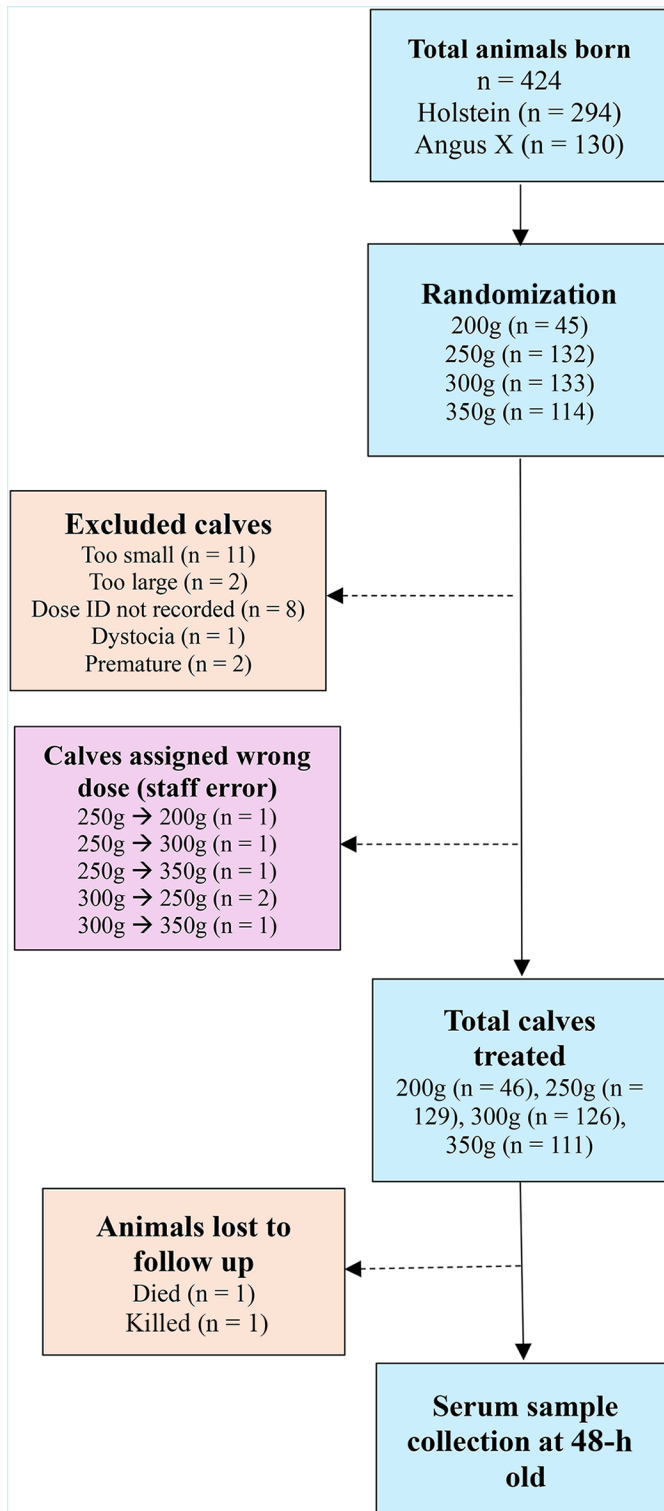
Total supply of IgG was calculated by summing the masses of IgG in each batch, which were calculated by multiplying the batch volume and concentration (tested using RID). The maximum dose that could be fed to the calves born during the observation period, when considering the available supply of IgG and the constraints of administration (maximum of 4 L of colostrum per calf using the available IgG concentrations), was termed the “maximum practically feasible dose.”

### Statistical Analysis

**Effect of IgG Dose on Serum IgG.** Analysis was conducted in R (version 4.5.1, R Core Team, 2018). An analysis log can be found at Hapukotuwa and Rowe (2026). The primary outcome variable was serum IgG concentration (g/L as determined by RID), which was used to determine the biologically optimal dose of colostral IgG by comparing serum IgG in calves administered 200, 250, 300, and 350 g oral IgG. An “as-treated” analytical approach was used, where subject treatment group was determined by the dose of colostrum administered, rather than the dose that was originally assigned during the randomization process. Figure 1 shows a causal diagram

(directed acyclic graph) that outlines our understanding of factors impacting serum IgG on commercial farms currently (panel A) and within the context of the experimental design in our study (panel B). Note that panel A is not intended to provide a complete understanding of all factors impacting serum IgG. The causal diagram in panel B was used in selection of covariates for control of confounding bias (VanderWeele et al., 2021). The final model to estimate the causal effect of the treatment (IgG dose) group (categorical variable) on serum IgG was a multivariable linear model that included colostrum volume as a covariate. Colostrum concentration was not included because it was highly correlated with volume, causing multicollinearity (identified using variance inflation factor, **VIF**). Other variables (time to feeding, calf sex, and breed) were not included as covariates because our causal diagram (panel B) did not identify biasing pathways that necessitated their inclusion.

**Investigating Colostrum Volume and Concentration, Calf Breed, and Time to Feeding.** As a secondary analysis, we investigated the importance of colostrum volume and concentration, calf breed, and time to first feeding. This was done by investigating effect modification for the relationship between IgG dose and serum IgG concentration (for all variables) and by constructing main effects models. Effect modification was evaluated by creating linear models with an interaction term between the potential effect modifier (i.e., breed, time to feeding, and volume fed) and treatment group. A *P*-value was calculated for the interaction term using a Type-2 analysis of deviance. If the *P*-value was  $\leq 0.05$ , then the null hypothesis (i.e., no effect modification) was rejected,



**Figure 2.** Description of enrollments, exclusions, and losses to follow-up during each stage of a randomized clinical trial evaluating the effect of oral colostrum IgG dose on serum IgG assessed at 2 d of age in Holstein and Angus × Holstein calves.

and a stratified model was reported. Effect modification was also evaluated by comparing crude and estimated marginal means for strata of the potential effect modifier. Models were stratified by breed to allow readers to generalize findings to their specific context, thereby improving the external validity of the current work.

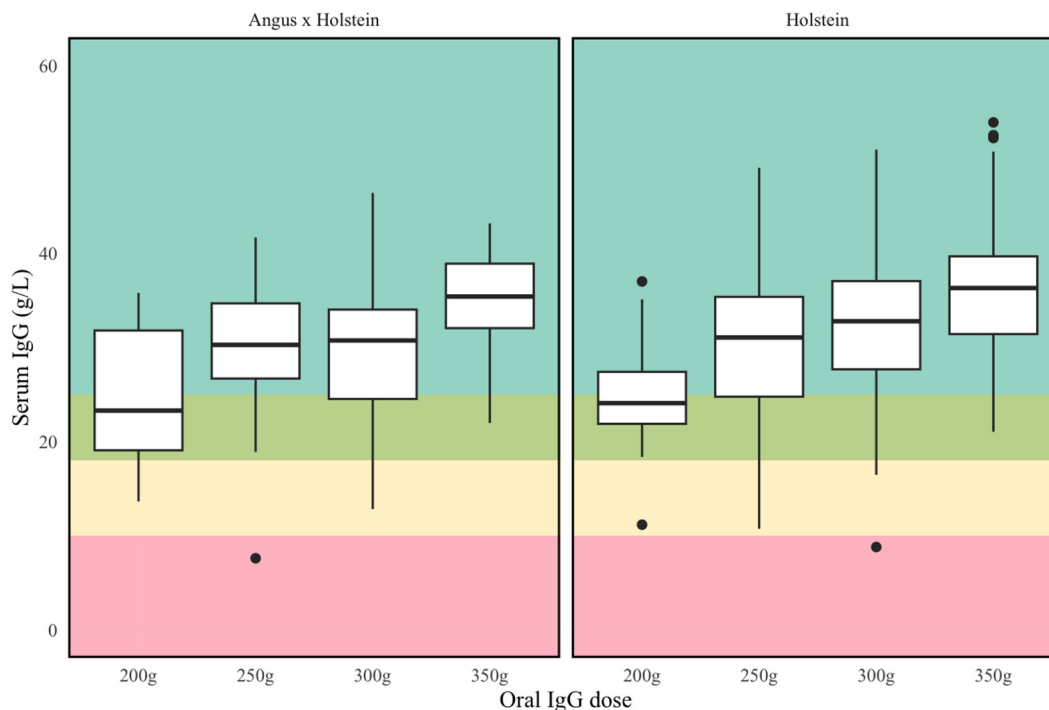
Colostrum volume, colostrum concentration, and time to feeding were evaluated as independent factors that could impact IgG absorption (main effects models). Univariable linear models were initially used to estimate the overall association between the variable of interest and serum IgG concentration (i.e., the combined effects mediated via IgG dose and IgG absorption efficiency). For the final models for volume and concentration, we included colostrum IgG dose as a covariate to estimate their effects on IgG that was mediated through IgG absorption efficiency only (Figure 1). We chose to conduct analysis this way because we judged the effect of volume and concentration on serum IgG via their effects on IgG dose to be self-evident (i.e., mass = volume × concentration). Investigations into the main effect of breed on serum IgG concentration were evaluated in an associated study (Hapukotuwa et al., 2026).

**Evaluation of Model Fit.** Sensitivity analyses were conducted to evaluate for potential selection bias (from using an “as-treated” analytical approach), confounding (calf breed, calf sex, and time to feeding), and violation of the independence assumption (autocorrelation of subjects within colostrum batches). Sensitivity analyses (Hapukotuwa and Rowe, 2026) indicated that effect estimates were similar using alternative analytical approaches (i.e., intention to treat analysis, adjusting for potential confounders, and using mixed models to account for autocorrelation). The independence assumption was assessed using a mixed-effects model to estimate the intraclass correlation coefficient. Normality of residuals was assessed using a quantile-quantile plot. To assess homoscedasticity, residuals were plotted against fitted values to check for patterns indicating nonconstant variance; no funneling or systematic structure was observed. The linearity assumption was evaluated by comparing distribution of data points to the fitted regression line. The VIF was calculated to assess the model for multicollinearity.

## RESULTS

### Descriptive Results

**Excluded Individuals.** Calves that were excluded from the study (and subsequently the final dataset) were for reasons such as dystocia (n = 1; 250 g n = 1), BW being outside of the inclusion criteria (visual estimate being <35 kg or >45 kg; “too small” [n = 11; 200 g n = 1; 250 g n =



**Figure 3.** Boxplot outlining raw serum IgG values at 2 d of age in calves randomized to receive 200 g ( $n = 45$ ), 250 g ( $n = 125$ ), 300 g ( $n = 122$ ), or 350 g ( $n = 107$ ) of colostral IgG within 3 h of birth. Each box represents the interquartile range (IQR), with the line inside the box representing the median data value. The vertical whisker lines extend to data points less than  $1.5 \times \text{IQR}$  from the edge of the box. Dots beyond the vertical lines indicate outlier values. Results are stratified by breed (Holstein and Angus  $\times$  Holstein). The background colors denote proposed serum IgG categories for passive immunity (Godden et al., 2019): excellent  $\geq 25$  g/L (teal), good 18 to  $<25$  g/L (olive), fair 10 to  $<18$  g/L (yellow), and poor  $<10$  g/L (pink).

3; 300 g  $n = 2$ ; 350 g  $n = 5$ ]; “too large” [ $n = 2$ ; 300 g  $n = 1$ ; 350 g  $n = 1$ ], and being premature ( $n = 2$ ; 300 g  $n = 2$ ).

**Missing Values.** Missing data pertained to staff not recording dose ID administered ( $n = 8$ ; 250 g  $n = 2$ ; 300 g  $n = 3$ ; 350 g  $n = 3$ ). Other missing values included calves that died ( $n = 1$ ) or were euthanized ( $n = 1$ ) and were coded as “lost to follow up,” as serum could not be collected at 2 d of life for IgG determination. Both the exclusions and missing data were evenly distributed across treatment groups.

**Enrollment.** Descriptive statistics for enrolled calves are outlined in Table 1. Figure 2 outlines the numbers of calves born, randomized, excluded, reassigned, and lost to follow-up during the trial period. A total of 424 Holstein and Angus  $\times$  calves were initially randomized into 4 treatment groups. The total number of calves in the study was higher than the intended sample size calculation because Angus  $\times$  calves were included and because a controlled modification was made in the treatment groups during the first week of the study (outlined in the “Materials and Methods” section). During the trial, 6 calves were incorrectly fed a dose intended for a different treatment group. The treatment group for these calves was reassigned to match the dose that was administered (“as-treated” analysis). Consequently, the final number

of calves included in the analysis was 45 for the 200 g of IgG group, 125 for the 250 g of IgG group, 122 for the 300 g of IgG group, and 107 for the 350 g of IgG group, resulting in a total of 399 calves.

#### Effect of Oral IgG Dose on Serum IgG Concentration

We found a dose-dependent effect of oral IgG dose on serum IgG concentration, which is shown in the box and whisker plot in Figure 3. Table 2 outlines the findings from multivariable linear models estimating the effect of oral IgG dose on serum IgG, adjusting for colostrum dose volume. Estimates were reported stratified by breed and for all breeds combined. In those models, the 250 g dose group was used as the reference, as the 200 g dose group had a small sample size and was only randomized to calves at the start of the study. There was no evidence for a dose of limiting returns at administered doses (200–350 g), as substantial increases in average serum IgG were seen for each 50 g increment in IgG dose. For example, when evaluating all breeds, the difference in mean serum IgG concentrations between the 250 g and 200 g groups was 5.6 g/L (95% CI: 2.0 to 9.1,  $P < 0.005$ ), between 300 g and 250 g groups was 2.8 (95% CI: 0.1 to 5.6,  $P = 0.04$ ), and between 350 g and 300 g groups was 4.8 (95%



**Table 2.** Linear regression estimates for the effect of colostral IgG dose (randomized to 200, 250, 300, and 350 g) on serum IgG at 2 d of age<sup>1</sup>

| Item                    | Model predicted IgG (g/L) | Difference (g/L IgG) | 95% CI       | P-value |
|-------------------------|---------------------------|----------------------|--------------|---------|
| All calves              |                           |                      |              |         |
| 200 g                   | 23.7 (21.0 to 26.4)       | −5.6                 | −8.2 to −2.9 | <0.005  |
| 250 g                   | 29.2 (27.8 to 30.6)       | Referent             |              |         |
| 300 g                   | 32.1 (30.7 to 33.4)       | 2.8                  | 0.7 to 4.9   | 0.04    |
| 350 g                   | 36.9 (35.2 to 38.5)       | 7.6                  | 5.2 to 10.1  | <0.005  |
| Holstein calves         |                           |                      |              |         |
| 200 g                   | 22.9 (19.2 to 26.6)       | −6.0                 | −9.7 to −2.4 | <0.005  |
| 250 g                   | 29.0 (27.2 to 30.7)       | Referent             |              |         |
| 300 g                   | 32.9 (31.2 to 34.5)       | 3.9                  | 1.4 to 6.4   | <0.005  |
| 350 g                   | 37.4 (35.5 to 39.4)       | 8.5                  | 5.5 to 11.4  | <0.005  |
| Angus × Holstein calves |                           |                      |              |         |
| 200 g                   | 25.4 (21.4 to 29.4)       | −4.6                 | −8.6 to −0.6 | 0.03    |
| 250 g                   | 30.0 (27.5 to 32.5)       | Referent             |              |         |
| 300 g                   | 30.0 (27.6 to 32.5)       | 0.0                  | −3.8 to 3.9  | 0.99    |
| 350 g                   | 35.0 (31.7 to 38.3)       | 5.0                  | 0.4 to 9.6   | 0.03    |

<sup>1</sup>Estimates were adjusted for colostrum dose volume. Results are stratified by breed (Holstein and Angus × Holstein).

CI: 2.3 to 7.3,  $P < 0.005$ ). The proportion of calves in the highest passive immunity category ( $\geq 25$  g/L) proposed by Godden et al. (2019) for serum IgG also increased with oral IgG dose (Figure 4).

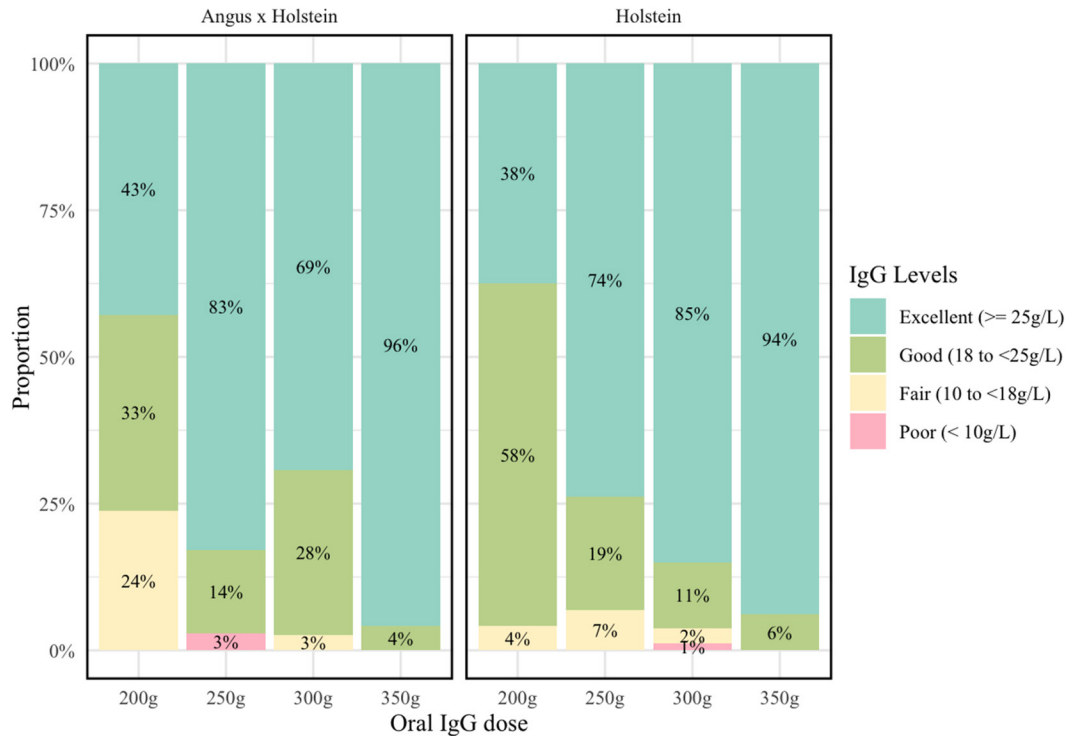
## Secondary Analyses

**Investigating Colostrum Volume and Concentration, Calf Breed, and Time to Feeding.** There was no evidence to suggest that the effect of IgG dose on serum IgG was modified by other variables. For example, the Type-2  $P$ -values for interaction terms between treatment group and breed ( $P = 0.6$ ), time to feeding ( $P = 0.9$ ), and volume ( $P = 0.3$ ) were all greater than  $P = 0.05$ . A more detailed description of these exploratory analyses can be found in the analysis log (Hapukotuwa and Rowe, 2026). Univariable main effect models found that volume (+4.2 g/L of serum IgG per increment in L of colostrum, 95% CI: 2.8 to 5.5,  $P < 0.005$ ) and concentration (+0.11 g/L of serum IgG per increment in g/L of IgG in colostrum, 95% CI: 0.04 to 0.18,  $P < 0.005$ ) were positively associated with serum IgG in univariable models. However, after controlling for IgG dose, the associations and 95% CI moved closer to 0 for both volume (−1.4, 95% CI: −3.5 to 0.6,  $P = 0.18$ ) and concentration (0.04, 95% CI: −0.02 to 0.1,  $P = 0.22$ ). This is illustrated as scatterplots for raw data in Figures 5, 6, and 7, which graphically show the relationship between each variable (volume, concentration, and time to feeding) and serum IgG after stratifying by colostrum dose. Serum IgG decreased by 0.2 g/L for each 10-min increment in time to feeding, but the 95% CI indicated that the estimate was imprecise (−0.04 to 0.5 g/L reduction per 10 min,  $P = 0.67$ ).

**Maximum Practically Feasible Dose.** A secondary study objective was to explore how IgG supply (colostrum volume and IgG concentration) impacted the practical feasibility of feeding IgG dose targets to the calves born over a 16-d observation period, when considering the constraints of administration (maximum of 4 L of colostrum per calf at a fixed IgG concentration). We found that the most important drivers of practical feasibility were the yield of IgG per calving, which was driven by the volume, and the IgG concentration of colostrum batches. A total of 117.9 kg of IgG was harvested over the 16-d period, which equates to 320 g per calf if divided equally among the calves born over that time period ( $n = 368$ ). Daily supply and demand for IgG fluctuated significantly, such that 130 to 635 g were available per calf if only colostrum harvested on the day of birth was available (Figure 8, panel A). There was also significant variability in IgG concentration in batches such that the maximum allowable dose ranged from 282 to 482 g/calf when feeding a maximum of 4 L (Figure 8, panel B). However, most (89%) colostrum batches were sufficiently concentrated to deliver 320 g in doses of  $\leq 4$  L. Based on these factors, we concluded that 320 g was the maximum practically feasible dose for the farm during the study period.

## DISCUSSION

Our objectives were to determine the biologically optimal dose of maternal colostral IgG by comparing serum IgG in calves administered 200, 250, 300, and 350 g oral IgG and evaluate the practical feasibility of various dose targets when considering the dynamics of IgG supply and demand. We found that there was a dose-dependent



**Figure 4.** Stacked bar chart showing the raw proportion of calves by treatment group of either 200, 250, 300, or 350 g of colostral IgG within 3 h of birth that fell into the proposed (Godden et al., 2019) serum IgG categories for passive immunity in a randomized clinical trial. The proposed passive immunity categories are excellent  $\geq 25$  g/L (teal), good 18 to  $<25$  g/L (olive), fair 10 to  $<18$  g/L (yellow), and poor  $<10$  g/L (pink).

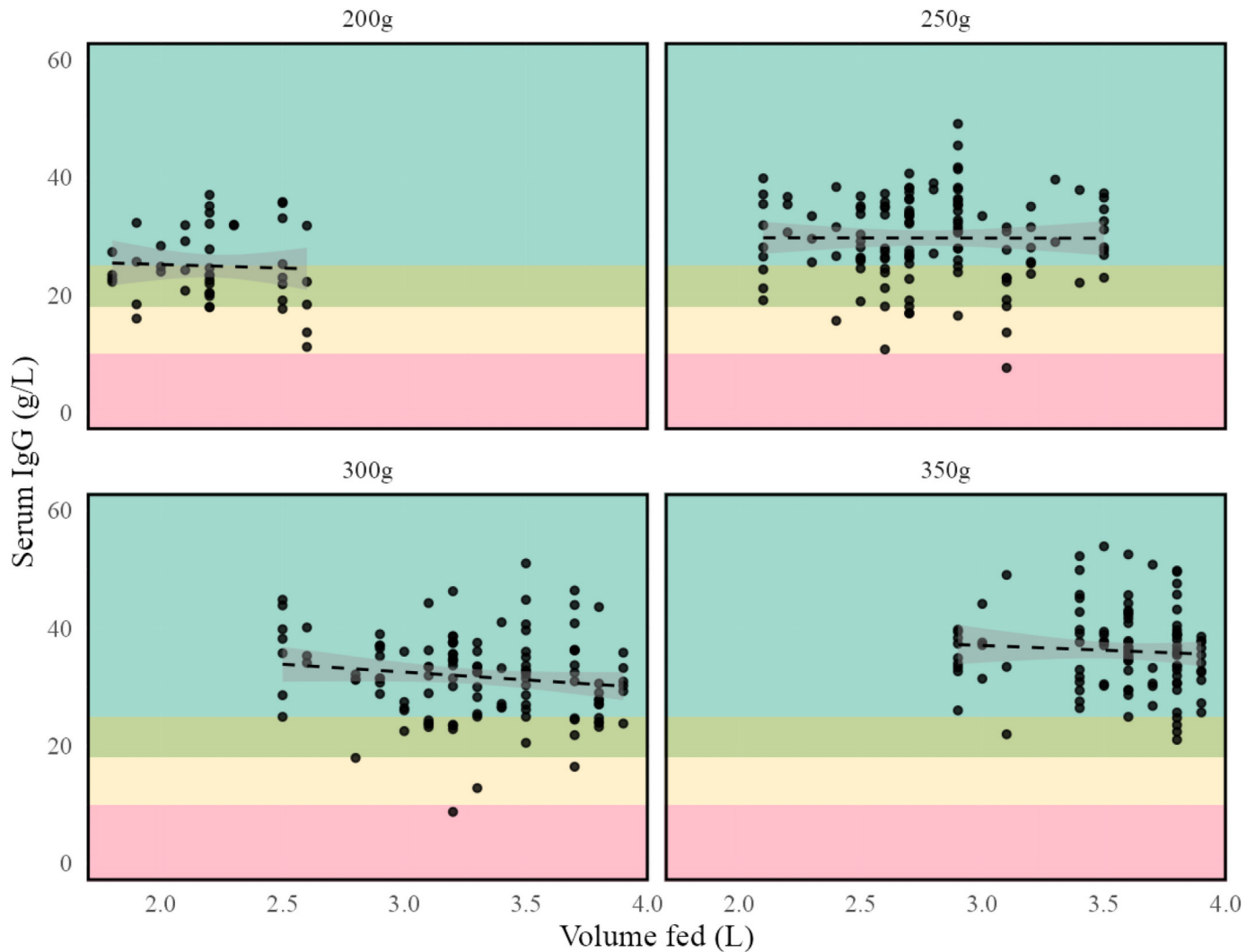
effect of oral IgG on serum IgG between 200 and 350 g of IgG that was not modified by colostrum volume and IgG concentration, time to feeding (if fed within 3 h of birth), or breed. We found that the IgG yield per cow and colostrum concentration were the most important factors limiting the maximum practically feasible dose and that under the conditions of this study (constrained to a single 4 L dose), 320 g was the maximum practically feasible dose. However, for producers to practically achieve maximal and consistent use of available colostrum, they require colostrum storage capacity.

### Dose Targets for Colostral IgG

We recommend that producers aim to feed calves as much colostral IgG as is practically feasible within 2 to 6 h of birth via either esophageal tube or teat feeder. This recommendation is supported by the finding that biologically meaningful, linear increases in serum IgG were achieved when increasing oral IgG doses (200 to 350 g), and previous studies demonstrated similar dose responses (Hopkins and Quigley, 1997; Godden et al., 2009; Osaka et al., 2014). Previously recommended oral IgG doses for calves ranged from 100 to 300 g per calf in a single feed (Besser et al., 1991; McGuirk and Collins, 2004; Chigerwe et al., 2008b; Godden et al., 2019; Lom-

bard et al., 2020). In our opinion, the use of fixed recommendations such as these is potentially misleading, as they implicitly suggest a diminishing marginal return as doses approach or exceed these thresholds. For instance, a target dose of 200 g could imply a greater incremental benefit in serum IgG when increasing the dose from 150 g to 200 g than from 200 to 250 g. Consequently, producers following such a recommendation may fail to realize the health benefits of higher doses.

It should be noted that our study only evaluated doses ranging from 200 to 350 g in a herd where mean calf birthweight was 39.6 kg (SD = 6.01). Therefore, it is plausible that a curvilinear relationship between oral IgG dose and serum IgG may exist in certain situations (e.g., doses  $>350$  g or smaller calves). This theory is supported by studies showing negative associations between IgG dose (or colostrum volume and concentration, both of which determine IgG dose) and apparent efficiency of absorption (Stott and Fellah, 1983; Besser et al., 1991; Frederick et al., 2025). If IgG absorption efficiency does decrease at higher IgG doses, then it is mathematically expected that the relationship between oral IgG and serum IgG will be curvilinear (IgG absorbed = dose  $\times$  efficiency of absorption). A limitation of the current study in evaluating this hypothesis comes from the evaluation of absolute IgG doses in grams per

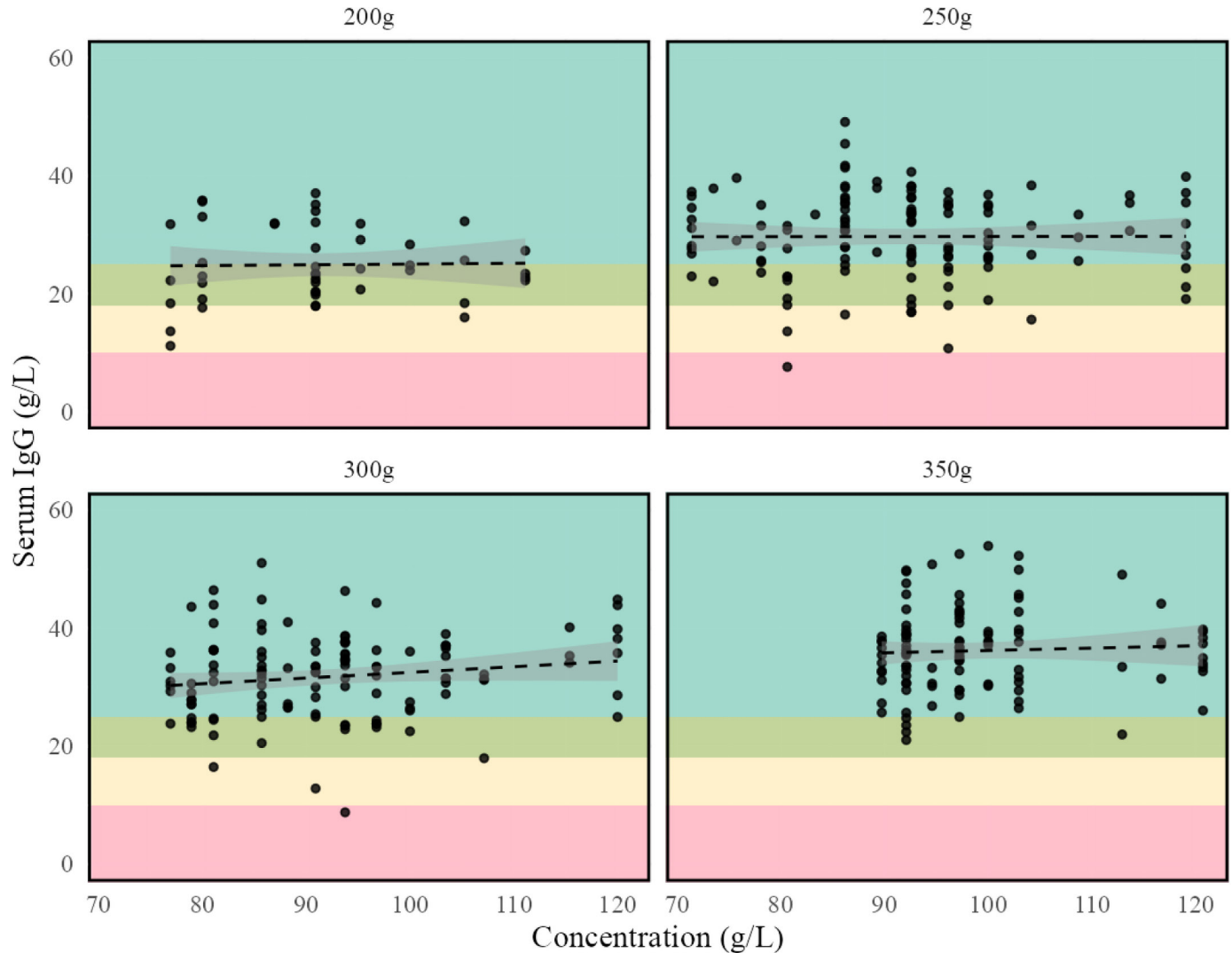


**Figure 5.** Scatter graphs and linear regression lines (dashed lines; gray shading indicates SE) of raw data showing the relationship between colostrum volume and serum IgG concentration (g/L) in calves administered 200, 250, 300, and 350 g of IgG within 3 h of birth. A multivariable linear model did not find an association between volume and serum IgG after controlling for IgG dose (estimate =  $-1.4$ , 95% CI:  $-3.5, 0.6$ ). The background colors denote proposed serum IgG categories for passive immunity (Godden et al., 2019): excellent  $\geq 25$  g/L (teal), good 18 to  $<25$  g/L (olive), fair 10 to  $<18$  g/L (yellow), and poor  $<10$  g/L (pink).

calf and not accounting for variation in calf birthweight. The estimated dose response, therefore, represents a population-averaged association across the birthweight distribution present in this herd (approximate mean = 39.6, SD = 6.01 kg), rather than a response conditional on measured individual calf BW. While this approach is directly relevant to farm level practice, where colostrum is typically administered per calf rather than per kilogram, future studies designed specifically to evaluate potential curvilinear relationships should include a broader range of IgG doses and explicitly account for individual calf birthweight, either by administering IgG as a dose rate in grams per kilogram or by including liveweight in statistical models.

### Other Drivers of Serum IgG Concentration

**Colostrum Volume and Concentration.** Previous studies have concluded that colostrum volume and concentration may influence IgG concentration due to their effects on IgG dose and on IgG absorption efficiency (Stott and Fellah, 1983; Besser et al., 1991; Frederick et al., 2025). Findings from our study support the positive effect of these factors on serum IgG concentration, most likely due to their effect on IgG dose. We did not identify associations between these variables and IgG concentration after controlling for their effect on IgG dose (via multivariable models or by stratification, see Figures 5 and 6). Chigerwe et al. (2008b) observed no significant effect



**Figure 6.** Scatter graphs and linear regression lines (dashed lines; gray shading indicates SE) of raw data showing the relationship between colostrum concentration and serum IgG concentration (g/L) in calves administered 200, 250, 300, and 350 g of IgG within 3 h of birth. A multivariable linear model did not find evidence for a strong association between colostrum concentration and serum IgG after controlling for IgG dose (estimate = 0.04, 95% CI: -0.02 to 0.1,  $P = 0.22$ ). The background colors denote proposed serum IgG categories for passive immunity (Godden et al., 2019): excellent  $\geq 25$  g/L (teal), good 18 to  $<25$  g/L (olive), fair 10 to  $<18$  g/L (yellow), and poor  $<10$  g/L (pink).

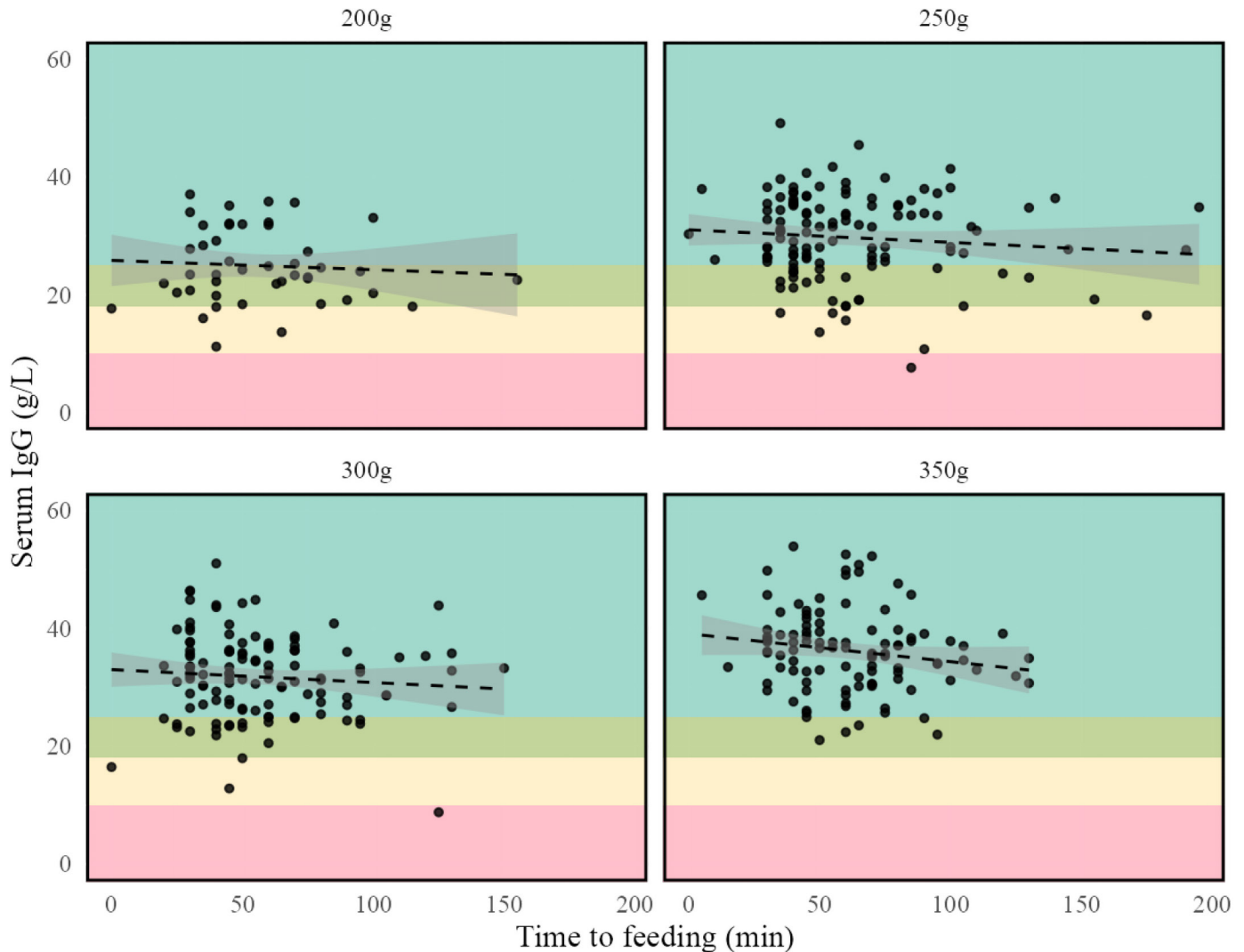
of volume of colostrum fed on serum IgG when administering colostrum doses  $\geq 150$  g of IgG. We hypothesize that our results are similar due to using relatively high doses of IgG (minimum 200 g) and the prompt timing of administration (within 3 h of birth) such that, if there was delayed emptying of IgG into the abomasum, the intestine would still have the high capacity to absorb the IgG delivered. In summary, we recommend that producers increase the volume and concentration of colostrum fed to maximize the total IgG dose, while considering practical factors such as calf comfort and maternal IgG supply.

**Time to Feeding.** The time lag from birth to feeding is generally understood to be the most important factor to impact IgG absorption (Stott et al., 1979). Findings from

main effect and interaction models in our study indicate that this factor is unlikely to play a substantial role in determining serum IgG when the time to feeding is  $<200$  min (i.e., promptly after birth) as occurred in the current study (Figure 7). We conclude that farmworkers should aim to feed colostrum as soon as possible after birth, as it has been demonstrated to be important elsewhere.

**Calf Breed.** Our finding that the effect of oral IgG and serum IgG is not modified by breed indicates that colostrum dosing for Angus X animals should follow the same recommendations for Holstein calves. A detailed analysis of the relationship between breed and serum IgG for calves enrolled in the current study can be found at Hapukotuwa et al. (2026). In that analysis, we found





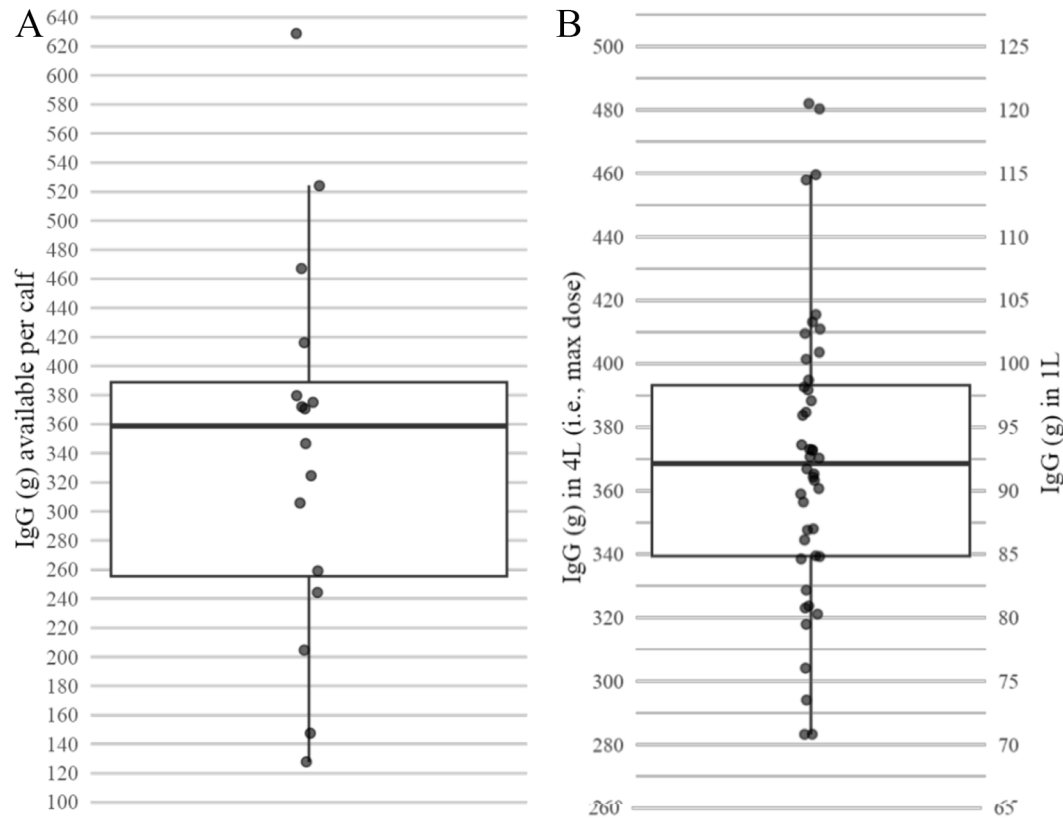
**Figure 7.** Scatter graphs and linear regression lines (dashed lines; gray shading indicates SE) of raw data showing the relationship between time from birth to feeding and serum IgG concentration (g/L) in calves administered 200, 250, 300, and 350 g of IgG within 3 h of birth. A univariable linear model estimated that serum IgG decreased by 0.2 g/L for each 10-min increment in time to feeding, but the 95% CI indicated that the estimate was imprecise (−0.04 to 0.5 g/L reduction per 10 min). The background colors denote proposed serum IgG categories for passive immunity (Godden et al., 2019): excellent  $\geq 25$  g/L (teal), good 18 to  $<25$  g/L (olive), fair 10 to  $<18$  g/L (yellow), and poor  $<10$  g/L (pink).

that serum IgG concentration was not associated with breed, but that serum TS (a commonly used proxy for serum IgG concentration) were lower in Angus X calves, thus requiring different cut points for using serum TS to monitor passive transfer in crossbred calves.

### **The Practically Feasible IgG Dose**

Given that no dose–response plateau was identified at IgG doses below 350 g, the practical implication is that serum IgG continues to increase with higher doses across this range and potentially at values  $>350$  g. Therefore, producers should aim to feed the maximum practically feasible dose for their farm by maximizing the total IgG

mass delivered while balancing this against logistical constraints (colostrum IgG concentration, colostrum volume harvested, maximum 10% BW volume/feed, and the frequency of delivery). Analysis of colostrum harvest data in our study indicated that colostral IgG yield was a limiting factor to maximizing IgG delivery. Therefore, producers can optimize IgG dosing by focusing on maximizing the IgG yield per calving. In the study herd, the maximum possible dose during the trial period was 320 g per calf. However, there was significant variation in daily IgG supply (130 to 635 g per calf) when calves were limited to colostrum harvested on the day of birth. This highlights the importance of colostrum storage to buffer day-to-day fluctuations regardless of the range



**Figure 8.** Panel A (left) shows a boxplot of oral IgG available per calf on each day ( $n = 16$ ) of a randomized clinical trial in calves receiving 200, 250, 300, or 350 g of colostrum IgG within 3 h of birth. Each dot represents the total IgG harvested from colostrum on each day of the study, divided by the number of calves born on that day. Panel B (right) shows a boxplot for the maximum oral IgG dose possible (primary y-axis), given the concentration (secondary y-axis) available in each colostrum batch ( $n = 50$ ). Each dot in the figure represents the different colostrum batches harvested during the trial. Each box represents the interquartile range (IQR), with the line inside the box representing the median data value. The vertical whisker lines extend to data points less than  $1.5 \times \text{IQR}$  from the edge of the box. Dots beyond the vertical lines indicate outlier values.

of IgG harvested on each individual farm. Additional methods used to increase the total mass of IgG harvested may include harvesting colostrum produced at the cow's second milking (Silva-del-Río et al., 2017; Kessler et al., 2020), as well as the addition of colostrum supplements (Quigley et al., 1998; Arthington et al., 2000; Dezfouli et al., 2007). It is important to note that this dose is not a blanket recommendation for all farms, as the dynamics of IgG supply, demand, and delivery vary significantly across seasons and farms.

Concentration is also an important limiting factor for IgG dosing because most calves are limited to volumes of 10% to 12% of BW due to abomasal capacity (Hamersma, 2011) and the pain observed at high volumes (Frederick et al., 2025). The range in concentration of batches in our study (71 to 121 g/L) corresponds to doses ranging from 282 to 482 g of IgG per calf when a maximum of 4 L is fed. The IgG concentration observed in our study is higher and less variable than what has been observed in other studies, reporting ranges of <10 to 120 g/L (Morin et al., 2010), 2 to 116 g/L (Morrill et al., 2012), 22.4 to

196.9 g/L (Beilmann et al., 2010), and 7.1 to 159 g/L (Quigley et al., 2013). The wide ranges of colostrum concentration in other studies compared with our own are likely due to the variability in timing of colostrum harvest because colostrum IgG concentration declines with time post calving (Moore et al., 2005; Morin et al., 2010; Conneely et al., 2013; Phipps et al., 2017). Other factors impacting IgG concentration include breed, parity, dry period length, periparturient nutrition, season, preharvest colostrum leakage, and total colostrum volume (Pritchett et al., 1991; Conneely et al., 2013; Phipps et al., 2017; Godden et al., 2019). It should also be noted that pooling of colostrum can negatively affect overall colostrum IgG concentration if high-volume, low-concentration harvests are included in pools (Pritchett et al., 1991; Weaver et al., 2000; Morrill et al., 2012) and can serve as a vehicle for dissemination of infectious agents shed in colostrum (Morrill et al., 2012).

The colostrum protocol in our study aimed to maximize serum IgG in all calves, including bulls and dairy beef cross animals. Dairy beef is becoming increasingly popu-

lar for the sustainable use of surplus calves (Berry, 2021; Rutherford et al., 2021; Renaud and Pardon, 2022). However, the presence of these animals on-farm increases the demand for colostrum IgG that has not been previously accounted for in colostrum management protocols with research being based primarily on Holstein heifer calves (Furman-Fratczak et al., 2011; Shivley et al., 2018; Urie et al., 2018b; Lombard et al., 2020). Shivley et al. (2019) conducted a survey to assess calf rearing practices for both bulls and heifers across the United States and found that 96.3% of bull calves received colostrum, and of that, 95.4% were hand-fed, compared with 100% of heifer calves being hand-fed colostrum. Their research also reports that bull calves were fed later and received an overall lower volume of colostrum in the first 24 h. In a cross-sectional Australian study, dairy beef crossbred calves were more likely to experience agammaglobulinemia (serum total protein concentration  $<4$  g/dL) given their low priority compared with replacement Holstein Friesian heifer calves (Vogels et al., 2013). In contrast, a study in the Netherlands found no differences in time to first colostrum feeding or total amount of colostrum fed between bull and heifer calves, with majority of calves being fed within 2 h of age and fed a total of  $\geq 6$  L within the first 24 h (Robbers et al., 2021a). The findings from Robbers et al. (2021a) and our study demonstrate that it is possible and practical to provide high doses of IgG to both heifer and bull calves.

### Study Limitations

A potential source of bias in this study is the mid-trial modification to treatment allocations, which resulted in 2 sequential randomization periods: d 1 to 7 (200, 250, or 300 g) and d 8 to 18 (250, 300, or 350 g). As a result, calves enrolled earlier were more likely to receive lower IgG doses than those enrolled later. If time-varying factors influencing serum IgG (other than oral IgG dose) differed between these periods, confounding could have occurred. However, exploratory analyses did not identify meaningful differences between the 2 periods in measured environmental or colostrum-related factors. Mean colostrum IgG concentration was similar between d 1 to 7 and 8 to 18 (93.1 vs. 94.4 g/L;  $P = 0.30$ ), ambient temperatures were comparable, and no rainfall occurred during the study. Nevertheless, unmeasured time-related factors cannot be completely excluded. Therefore, period effects remain a potential, although unquantified, source of bias. Additionally, the use of an “as-treated” analytical approach could have potentially caused confounding bias, as it can counteract the beneficial effects of randomization. This was minimized by study subjects and farmworkers being blinded to treatment group and the fact that treatment switching was uncommon ( $n = 6$ ).

Another important methodological issue is the absence of individual calf BW measurements. Visual assessment of birthweight was used to assess eligibility for inclusion. Consequently, selection bias could have occurred if inclusion assessment differed across the IgG dose groups (e.g., smaller calves being more or less likely to be included in a particular group than another). We tried to prevent this situation from occurring by having all weight estimations being performed by one investigator (Dini Hapukotuwa), who was blinded to treatment group when estimating BW. Failure to account for variation in liveweights when estimating the effect of IgG dose on serum IgG response would be expected to reduce precision of the estimated dose response and restrict interpretation to a population-averaged effect rather than a weight-adjusted association. Consequently, future studies should account for liveweight either by allocating dose rates in grams of IgG per kilogram of BW or by including liveweight in statistical models. Estimation of a dose response curve expressed in grams per kilogram would likely enhance external validity by allowing findings to be more readily generalized across herds with different calf size distributions. In contrast, the present results reflect the average response observed in a population with a mean birthweight of 39.6 kg (SD = 6.01). Producers should therefore interpret the reported dose response within the context of calf size distributions similar to those represented in this study.

Serum samples were run in triplicate to measure IgG concentrations using RID. For each RID testing batch, standards were distributed across 3 plates in the batch to provide triplicate standard readings. Some studies (Quigley et al., 2013; Elsohaby and Keefe, 2015) describe running standards on every plate to reduce individual plate variation. In the present study, plate-level variance was not addressed explicitly in this manner, which may have reduced analytical precision. The magnitude of any resulting imprecision is unknown; however, previous work with this assay has reported minimal plate-to-plate variability (Thompson et al., 2023). Another limitation of this study was that non-IgG nutrient composition in colostrum could not be controlled. Therefore, it must be noted that IgG absorption may also be affected by varying compositions of nutrients in colostrum, such as fat, lactose, proteins, and oligosaccharides, which may affect absorption efficiency (Davenport et al., 2000). Furthermore, colostrum doses were sometimes frozen before feeding, and others were not frozen. Nevertheless, several studies have demonstrated that freezing has a negligible effect on IgG concentration (Balthazar et al., 2015; Morrill et al., 2015; Westhoff and Mann, 2025).

The generalizability of the findings is limited by the management context in which the study was conducted. Colostrum was harvested within 2 h of calving, pooled,

heat-treated, and fed within 3 h of birth. These conditions are not typical of many Australian and New Zealand pasture-based systems. For example, it is likely that the maximum practically feasible dose in such systems is less than 320 g due to later first postcalving harvest of colostrum resulting in lower IgG concentrations (Besser and Gay, 1985; Mann et al., 2025), which imposes a limit on dose if restricted to single 4 L feeding. Phipps et al. (2016) found that only 46.7% of samples met the current standard for “good quality” colostrum (>50 g/L IgG), consistent with another Australian study harvesting 52% of good quality samples (Chuck et al., 2017). In contrast, 100% of samples in our study were >50 g/L IgG, and intensively managed herds in the United States (where the calving to colostrum harvest is usually shorter) have reported higher proportions, including 68% (Chigerwe et al., 2008a), 77.3% (Urie et al., 2018a), and 83% (Beilmann et al., 2010).

Furthermore, the short duration of the trial further reduces the generalizability of our results, as factors such as seasonality could not be assessed. Nevertheless, the recommendation to identify a practically feasible dose that is personalized to the individual farm remains, as this will allow producers to account for seasonal changes in IgG supply.

Finally, the effect of oral IgG dose on morbidity, mortality, or productivity was beyond the scope of the current study. This was due to the sample size being underpowered for these outcomes. Future work with larger sample sizes per oral IgG treatment group could estimate effects on health and productivity.

## CONCLUSIONS

In a single commercial dairy herd where calves weighed 39.6 kg on average at birth, feeding increasing oral doses from 200 to 350 g of IgG in a single colostrum feed was associated with progressively higher average serum IgG concentrations, without evidence of a plateau within the evaluated range. These findings demonstrate a clear dose response under field conditions and indicate that increasing the overall IgG mass delivered per calf at first feeding yields measurable increases in serum IgG. Herd level analyses further demonstrated that colostrum IgG concentration and total IgG yield per cow (i.e., volume × concentration) were the primary determinants constraining IgG availability per calf. While the magnitude of the incremental increase in serum IgG concentration per increasing IgG dose administered may differ in herds with different calf size demographics, feeding protocols, or colostrum management systems, these results support maximizing oral IgG delivery per calf within individual farm practical constraints, rather than relying on a fixed universal target dosage.

## NOTES

The funding of this study was provided by the University of Sydney’s Hugh Hughes Scholarship for Combined Master’s Degree in Veterinary Studies and Veterinary Clinical Studies as well as by additional academic funding provided by supervisors John House and Sam Rowe (University of Sydney, Camden, NSW, Australia). Supplemental material for this article is available at [https://osf.io/suf4m/overview?view\\_only=ab00741c55a348d1a4b4054e6f0d1f41](https://osf.io/suf4m/overview?view_only=ab00741c55a348d1a4b4054e6f0d1f41) and [https://rpubs.com/samrowe/igg\\_dose\\_trial\\_2024](https://rpubs.com/samrowe/igg_dose_trial_2024) (analysis log). D. A. Hapukotuwa was involved in study conceptualization, fieldwork, laboratory testing, and manuscript preparation. J. K. House and K. S. Denholm were both involved in study conceptualization and manuscript editing. S. M. Rowe was the supervisor for this study and was involved in study conceptualization, data analysis, and manuscript editing. Ethics approval was granted by the University of Sydney Animal Ethics Committee (2024/AE002408) prior to the commencement of the study. The authors have not stated any conflicts of interest.

**Nonstandard abbreviations used:** Angus X = Angus × Holstein; IQR = interquartile range; RID = radial immunodiffusion; VIF = variance inflation factor.

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